

Stepper motor

DO

OUTPUT

MOVEMENT

What does it do?

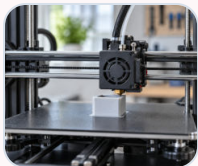
A stepper motor turns in small, repeatable steps instead of spinning freely.

Energising its coils in sequence moves the shaft one step at a time and holds its position.

The ULN2003 board supplies the coil current for the common 28BYJ-48 motor shown here.

Uses

3D printer positioning



DETECT
Read next print position

DECIDE
Calculate required steps

DO
Move the print head

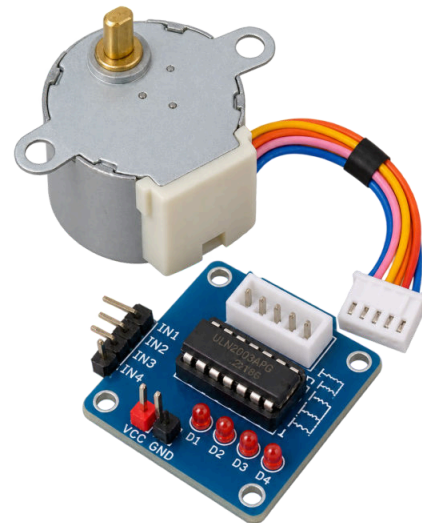
Snack lift



DETECT
Snack request arrives

DECIDE
Which floor needs it

DO
Step lift into place



Output shaft

Moves in measured steps

Five-wire plug

Connects to the driver

IN1 to IN4

Coil sequence

5V and GND

Driver power

Code

● Raspberry Pi

```
from gpiozero import OutputDevice
from time import sleep

coils = [OutputDevice(pin) for pin in (22, 27, 17, 4)]
sequence = ((1, 0, 0, 0), (0, 1, 0, 0),
            (0, 0, 1, 0), (0, 0, 0, 1))

for pattern in sequence:
    for coil, value in zip(coils, pattern):
        coil.value = value
    sleep(0.002)
```

● Raspberry Pi Pico

```
from picozero import Stepper

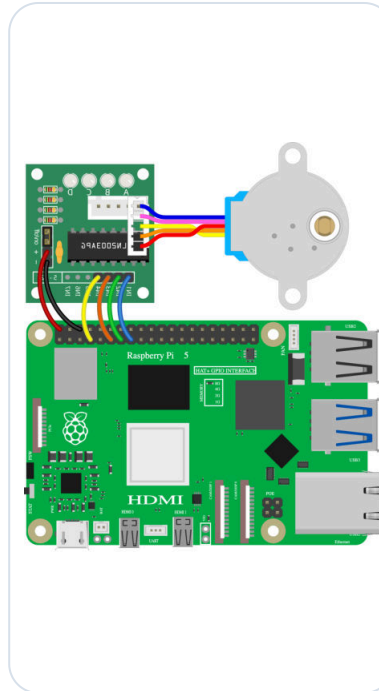
motor = Stepper((16, 17, 18, 19))

motor.step(100)
```

Repeat the Pi sequence to keep moving.
Reverse either board's sequence to turn the other way.

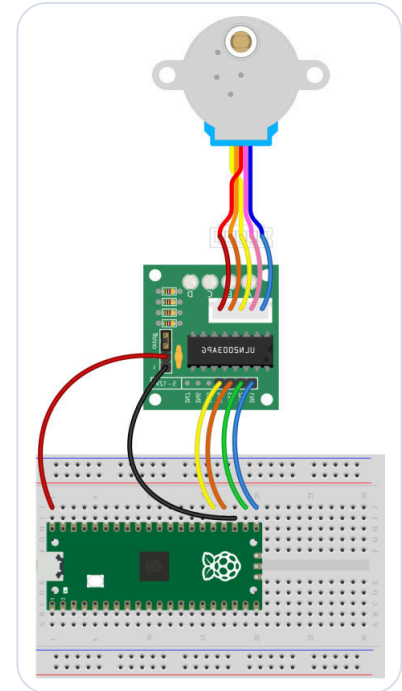
Wiring

RASPBERRY PI 5



IN1-IN4 → GPIO **22, 27, 17, 4**; driver →
5V and **GND**

PICO



IN1-IN4 → GP **16, 17, 18, 19**; driver →
VBUS and **GND**